

HARSHITH SIDDARDHA MANNE

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SUMMARY

Cloud and DevOps engineer with hands-on experience architecting production AWS infrastructure, automating deployments through Infrastructure as Code, and shipping containerized backend systems on ECS, EKS, and Kubernetes. Skilled in CI/CD automation, observability (Prometheus, Grafana, Loki, Datadog), and cost-efficient cloud-native design aligned with the AWS Well-Architected Framework. Extending these foundations into multi-tenant SaaS platforms and knowledge-graph-grounded AI systems.

EXPERIENCE

DEEP ALGORITHMS & SOLUTIONS

May 2023 – May 2024

Cloud / DevOps Engineer

Hyderabad, India

- Architected a highly available 3-tier AWS infrastructure with Terraform and CloudFormation across VPC, ALB, ECS, RDS, and S3, cutting provisioning errors by 40% and infrastructure setup time by 35% through standardized IaC modules.
- Containerized Python-Flask microservices and deployed to ECS Fargate and **Kubernetes (EKS)** with Helm-managed workloads, Horizontal Pod Autoscaling, and Nginx ingress, supporting 2x traffic growth without performance degradation.
- Engineered CI/CD pipelines with AWS CodePipeline, CodeBuild, and CodeCommit, reducing deployment failures by 30% and accelerating release cycles by 40% across dev and staging environments.
- Built observability with **Prometheus, Grafana, and Loki** for cluster metrics, application logs, and SLO dashboards; integrated CloudWatch alarms and Datadog-style golden-signal monitoring to cut incident MTTR by 35%.
- Optimized Docker multi-stage builds and ECS task definitions to cut deployment time by 45%; configured autoscaling, Multi-AZ RDS, and developed Lambda-based ML batch pipelines in private subnets using SSM, Secrets Manager, S3, and NAT Gateway for secure outbound access.

BACKFLIPT (XENOVIOUS, INC.)

Jan 2023 – Apr 2023

Associate Software Developer Intern

Hyderabad, India

- Designed high-concurrency RESTful services in **Java, Spring Boot, and Reactive Spring** for the International Data Corporation (IDC) project, improving API response efficiency by 30% under load.
- Integrated backend APIs with React / React Native clients using Redux state management, reducing frontend-backend integration defects by 25%; tuned and resolved production issues on AWS-hosted environments, cutting incident turnaround by 35%.

DEEP ALGORITHMS & SOLUTIONS

Jan 2021 – Mar 2022

DevOps & Backend Engineer Intern

Hyderabad, India

- Engineered RESTful APIs for an AI NLP chatbot using **Python (Flask, Django)** with JWT auth, request validation, and SQL/NoSQL persistence; containerized services with Docker and Docker Compose.
- Provisioned AWS via CloudFormation (EC2, S3, IAM, ALB) with autoscaling and lifecycle policies; integrated CloudWatch / CloudTrail for monitoring; built end-to-end CI/CD with AWS CodePipeline / CodeBuild / CodeCommit and PR-based branching across dev / stage / prod.

PROJECTS

AI-Driven AWS Cost Optimization Platform

- Built a multi-tenant SaaS that ingests live AWS topology, CloudWatch metrics, and Cost Explorer data, projects it onto a **12k-node / 14k-edge knowledge graph** (NetworkX MultiDiGraph) of AWS services, 104 optimization patterns, and cost factors, and surfaces ranked recommendations via a deterministic 5-stage solver; validated end-to-end via scenario-based regression harness.
- Implemented graph-grounded **RAG chat** on **Amazon Bedrock (Claude Sonnet)** with citation-required prompts and a heuristic-then-upgrade polling flow to operate within API Gateway's 30-second HTTP limit; built live AWS scanner using STS AssumeRole + boto3 across 12+ services (EC2, RDS, EKS, Lambda, DynamoDB, S3) feeding both the solver and a hierarchical SVG architecture view.
- Implemented multi-tenant **RBAC** (Postgres row-level isolation, JWT + refresh tokens, compliance filtering for SOC 2 / HIPAA / PCI-DSS); deployed FastAPI as containerized Lambdas via Terraform with GitHub Actions CI/CD.

TECHNICAL SKILLS

Cloud / DevOps / IaC: AWS (EC2, ECS, EKS, Lambda, S3, RDS, DynamoDB, VPC, IAM, ALB, API Gateway, CloudFront, KMS, Bedrock), Terraform, CloudFormation, Docker, **Kubernetes / EKS** (Helm, HPA), Nginx, CI/CD (GitHub Actions, AWS CodePipeline / CodeBuild)

Observability: Prometheus, Grafana, Loki, **Datadog**, CloudWatch (Logs, Metrics, Insights, Alarms), CloudTrail, distributed tracing, Golden-signal monitoring

Backend / AI Systems: Python (FastAPI, Flask, Django, Asyncio, SQLAlchemy), Java (Spring Boot, Reactive Spring), TypeScript, REST APIs, microservices, JWT/OAuth/RBAC

Data / AI / Knowledge Systems: PostgreSQL, MongoDB, Alembic, Amazon Bedrock, Anthropic SDK, RAG pipelines, agentic AI workflows, knowledge graphs (NetworkX), MCP, FinOps, AWS Well-Architected Framework

EDUCATION

Arizona State University, Tempe, AZ | Master of Science in Information Technology

May 2026

Jawaharlal Nehru Technological University (JNTU) Kakinada | Bachelor's in Computer Science

May 2023